

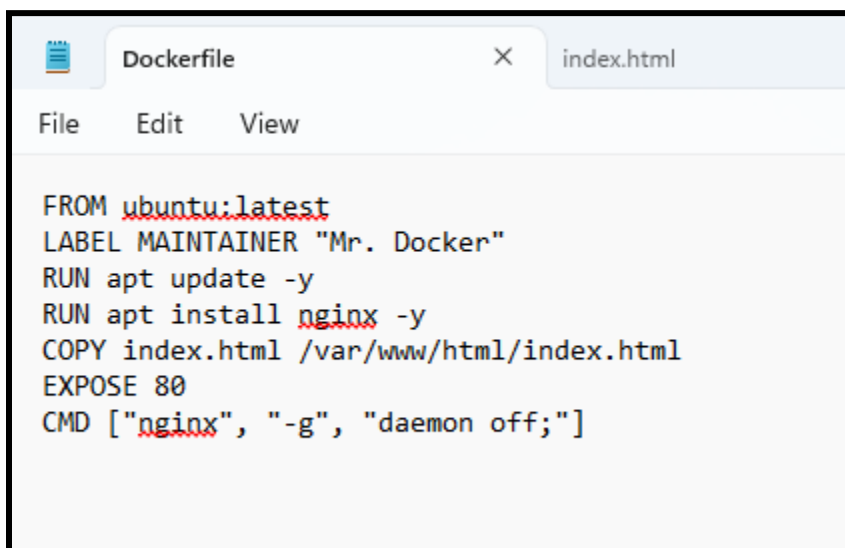
EXPERIMENT NO. : 10

Directory creation:

```
C:\Users\MOAD>mkdir rohini04  
C:\Users\MOAD>cd rohini04  
C:\Users\MOAD\rohini04>
```

Creation of docker file:

```
C:\Users\MOAD\rohini04>notepad Dockerfile|
```

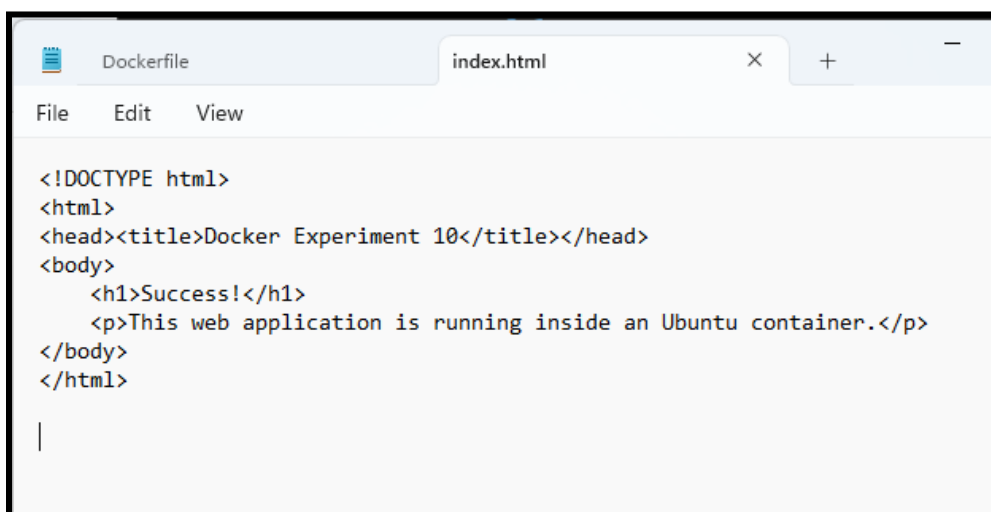


The screenshot shows a Notepad window with two tabs: 'Dockerfile' and 'index.html'. The 'Dockerfile' tab is active, displaying the following text:

```
FROM ubuntu:latest  
LABEL MAINTAINER "Mr. Docker"  
RUN apt update -y  
RUN apt install nginx -y  
COPY index.html /var/www/html/index.html  
EXPOSE 80  
CMD ["nginx", "-g", "daemon off;"]
```

Creation of html file:

```
C:\Users\MOAD\rohini04>notepad index.html_
```



The screenshot shows a Notepad window with two tabs: 'Dockerfile' and 'index.html'. The 'index.html' tab is active, displaying the following HTML code:

```
<!DOCTYPE html>  
<html>  
<head><title>Docker Experiment 10</title></head>  
<body>  
  <h1>Success!</h1>  
  <p>This web application is running inside an Ubuntu container.</p>  
</body>  
</html>
```

The cursor is positioned at the end of the first empty line below the HTML code.

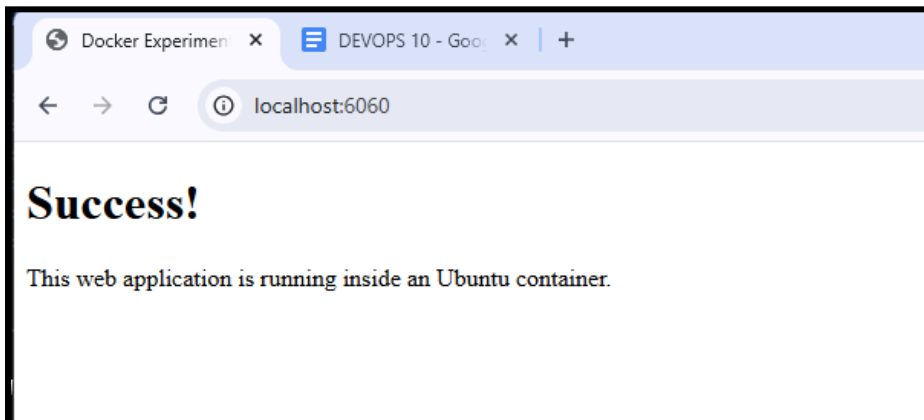
Building and running the web application on Docker Engine:

```
PS C:\Users\MOAD\Desktop\rohini04> docker build -t mynginx .
[+] Building 0.1s (8/8) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 220B                             0.0s
=> WARN: LegacyKeyValueFormat: "LABEL key=value" should be used instead of legacy "LABEL key value" format (line 2) 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/3] FROM docker.io/library/ubuntu:latest                  0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 233B                                  0.0s
=> CACHED [2/3] RUN apt update -y && apt install nginx -y      0.0s
=> CACHED [3/3] COPY index.html /var/www/html/index.html        0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:13ace45bc9ac66d459727c9f2b7479aa3acef0a5440e9c1f4ca2211ed2dacdc6 0.0s
=> => naming to docker.io/library/mynginx                      0.0s

1 warning found (use docker --debug to expand):
- LegacyKeyValueFormat: "LABEL key=value" should be used instead of legacy "LABEL key value" format (line 2)

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview
```

```
PS C:\Users\MOAD\Desktop\rohini04> docker run -d -p 6060:80 --name experiment-final mynginx
eb8c6a54f03b01de0aa18d700b53875eb1b0ed30c9f7784673a6f13f21e9e054
PS C:\Users\MOAD\Desktop\rohini04> |
```



Docker desktop :

Containers [Give feedback](#)

Container CPU usage ⓘ
0.00% / 1200% (12 CPUs available)

Container memory usage ⓘ
8.73MB / 7.46GB

Show charts

☒ Only show running containers

☐		Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	☐	Swara	102914b54a48	ubuntu		0%	4 hours ago	
☐	☐	pedantic_vaughan	7b2f3ea9edc6	ubuntu		0%	5 hours ago	
☐	☐	sharp_visvesvaraya	7633fb1f86af	my_valen_nginx:latest		0%	1 day ago	
☐	☐	admiring_hopper	d35aa1ba4bca	valen	8080:80	0%		
☒	☐	brave_cerf	e024fb678447	nginx		0%	5 months ago	
☐	☐	elegant_sammet	995692aad98	ubuntu:latest		0%	5 months ago	
☐	☐	happy_carver	7f11ca5967ae	nginx:latest		0%	5 months ago	
☐	☐	keegan	90f919807062	ubuntu		0%	5 months ago	
☐	☐	experiment-container	c3fddc8cdc79	mynginx	8080:80	0%		
☐	☒	experiment-final	eb8c6a54f03b	mynginx	6060:80	0%	8 minutes ago	

☐	ubuntu	jammy	28530540ca	1 month ago	77.57 MB				
☐	valen	latest	417e9ab1eca5	5 months ago	140.88 MB				
☐	nginx	latest	9d0e6f6199dc	5 months ago	151.86 MB				
☐	ubuntu	<none>	97bed23a3497	6 months ago	78.12 MB				
☐	mynginx	latest	13ace45bc9ac	41 minutes ag	145.76 MB				

Showing 16 items

Answer the following:

- What is a Dockerfile?

A Dockerfile is a text file that contains a set of instructions used to automatically build a Docker image. It defines the base image, application code, dependencies, and commands required to run the application inside a container.

- Explain Dockerfile commands with syntax and example.

Some commonly used Dockerfile commands are:

- **FROM**
Purpose: Specifies the base image
Syntax: FROM <image_name>
Example: FROM node:14
- **WORKDIR**
Purpose: Sets the working directory inside the container
Syntax: WORKDIR /path
Example: WORKDIR /app
- **COPY**
Purpose: Copies files from local system to container
Syntax: COPY <source> <destination>
Example: COPY . /app
- **RUN**
Purpose: Executes commands during image build
Syntax: RUN <command>
Example: RUN npm install
- **EXPOSE**
Purpose: Specifies the port to be used by container
Syntax: EXPOSE <port>
Example: EXPOSE 3000
- **CMD**
Purpose: Defines default command to run container
Syntax: CMD ["executable", "arg1"]
Example: CMD ["node", "app.js"]